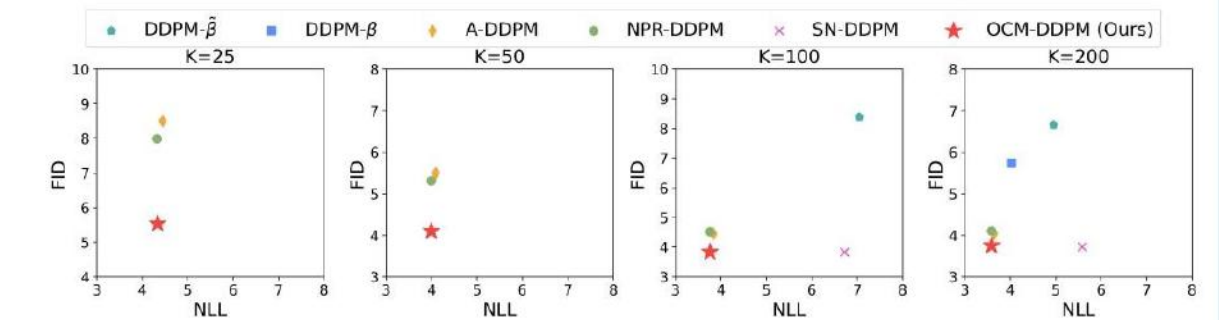


OPTIMAL DIAGONAL COVARIANCE MATCHING

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WHYNOW

- COVARIANCE = FIDELITY & STABILITY
- FIXED NOISE SCHEDULES MISALIGN
- BAD VARIANCE HURTS FID-NLL, STEPS
- OCM LEARNS NEEDED DIAGONAL COVAR



GOAL, USE CASE

- Learn and plug in the optimal x_t -dependent diagonal reverse covariance to boost sample quality and likelihood at minimal extra cost.
- Use cases: faster few-step generation; lower FID at fixed K; higher likelihood for selection; drop-in with DDPM/DDIM + classifier-free guidance.

DIFFUSION FOUNDATIONS

- Forward $q(x_t|x_0)$: add Gaussian noise step by step.
- Mean: $\sqrt{\alpha_{\bar{t}}}$ x_0 ; Cov: $(1-\alpha_{\bar{t}}) \mathbf{I}$.
- DDPM (Markovian) vs DDIM (non-Markovian) essentials.
- Reverse: $p_{\theta}(x_{t-1}|x_t)$; learn mean, cov.
- Score network sets the mean via $\text{grad log } q(x_t)$.
- Fixed covariance hurts quality, especially at small K.

OCM

$$\Sigma(\tilde{x}; \phi) = (\sigma^4 h_\phi(\tilde{x}) + \sigma^2 I) / \alpha^2. \tag{12}$$

- PREDICT DIAG COVARIANCE FROM X_T
- UNBIASED LOSS VIA SCORE-CURVATURE
- PLUG INTO DDPM/DDIM UPDATES
- BETTER FIDELITY AND STABILITY

CURV->VAR

Sharper score curvature -> higher uncertainty; via diag Hessian.
OCM uses Rademacher Hessian-vector probes for unbiased diagonals.

$$\text{diag}(H(\tilde{x})) \approx 1/M \sum_{m=1}^M v_m \odot H(\tilde{x}) v_m, \tag{9}$$

OCM-DPM SYSTEM

- Shared backbone: $s_\theta(x_t)$ for mean/score, plus covariance head $h_\phi(x_t)$. Train score first; then OCM learns diagonal Σ via unbiased objective.
- At sampling, μ from s_θ ; diagonal Σ from h_ϕ . Drop-in for existing DPMs. A compatible deterministic DDIM variant uses the same covariance head.

$$s_\theta(x_t) = \text{NN}_1(\text{BaseNet}(x_t, t; \theta_1); \theta_2), \quad h_\phi(x_t) = \text{NN}_2(\text{BaseNet}(x_t, t; \theta_1); \phi) \tag{20}$$

METHOD

$$L_{\text{ocm}}(\phi) = \mathbb{E}_{q(\tilde{x})p(v)} \|h_\phi(\tilde{x}) - v \odot H(\tilde{x})v\|_2^2, \tag{11}$$

- MOMENTS FROM SCORE + CURVATURE
- RAND HVPS YIELD UNBIASED DIAG
- AMORTIZED HEAD $H\Phi$, SIMPLE L2
- PLUG $\Sigma_{T-1}(X_T; \Phi)$ INTO REVERSE
- SHARE SCORE BACKBONE: LOW, STABLE

WHY OCM

- Directly matches the x_t -dependent diagonal of the reverse conditional $p(x_{t-1}|x_t)$, not schedule surrogates.
- Unbiased randomized probes tame second-order instability—no noisy curvature hacks or bias.
- x_t -dependent accuracy at modest compute and parameter cost; see comparison table.

How many network passes are required for estimating the covariance.

Covariance Type	+ #Passes	Intuition
x_t -independent Isotropic $\hat{\beta}$ (Ho et al., 2020)	0	Cov. of $q(x_t x_{t-1})$
x_t -independent Isotropic $\hat{\beta}$ (Ho et al., 2020)	0	Cov. of $q(x_t x_{t-1}, x_0)$
x_t -independent Isotropic Estimation (Bao et al., 2022b)	0	Estimate from data
x_t -dependent Diagonal VLB (Nichol & Dhariwal, 2021)	1	Learn from data
x_t -dependent Diagonal NS (Bao et al., 2022a)	1	Learn from data
x_t -dependent Diagonal OCM (Ours)	1	Learn from score
x_t -dependent Diagonal Estimation (Zhang et al., 2024b)	$2M$	Estimate from score
x_t -dependent Diagonal Analytic (Zhang et al., 2024b)	D	Calculate from score
x_t -dependent Full Analytic (Zhang et al., 2024b)	D	Calculate from score

TIME/COST

- Lightweight covariance head
- Minimal per-step compute
- Modest model size
- Small inference-time change

Overhead: a few percent	SCORE PREDICTION NETWORK	SCORE & SN PREDICTION NETWORKS	SCORE & DIAGONAL HESSIAN PREDICTION NETWORKS
Baseline sample efficiency kept	CIFAR10 (CS) 92.59 M / 44.37 MS	52.55 M / 44.61 MS (+0.5%)	52.55 M / 44.51 MS (+0.3%)
Quality gains: FID, likelihood	CIFAR10 (CS) 52.54 M / 45.13 MS	52.55 M / 45.24 MS (+0.2%)	52.55 M / 45.23 MS (+0.2%)
Near-unchanged runtime	CELEBA 64X64 78.70 M / 67.88 MS	78.71 M / 69.15 MS (+1.9%)	78.71 M / 68.89 MS (+1.5%)
	IMAGENET 64X64 121.06 M / 106.58 MS	121.49 M / 112.53 MS (+5.6%)	121.49 M / 112.23 MS (+5.3%)
	LSUN BEDROOM 113.67 M / 692.58 MS	114.04 M / 771.55 MS (+11.4%)	114.04 M / 770.73 MS (+11.2%)
	IMAGENET 256X256 675.13 M / 22.84 MS	MISSING MODEL	677.80 M / 23.01 MS (+0.7%)

GAINS

- OCM IMPROVES 2D TOY COVARIANCE
- LOWER FID AT SAME K, BROAD SETS
- $FID \approx 6$ ACHIEVED WITH FEWER STEPS
- GAINS PERSIST UNDER CFG GUIDANCE



RECAP

- OCM learns optimal diagonal reverse covariance and plugs into DDPM/DDIM to improve both sample fidelity and likelihood.
- It delivers these gains with minimal compute overhead and acts as a drop-in module for both training and sampling.
- We will release code and invite collaboration to integrate OCM into fast samplers and strong evaluation pipelines.
- BEYOND DIAGONAL: STRUCTURE
- REFINE SMALL-T + PROBE DESIGN
- SCALE: LATENT + DIT, 256×256
- INTEGRATE FAST SAMPLERS + EVALS